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Controller and method of operating the same

Abstract

A controller controls a semiconductor memory device including a plurality of pages. The controller includes a command generator configured to generate a command for controlling a program operation or a read operation of a semiconductor memory device, and a data recovery manager configured to generate parity data corresponding to a weak page among a plurality of pages included in the semiconductor memory device and recover data programmed to the weak page.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2016/0364181	12/2015	McGlaughlin	N/A	G06F 3/0629
2020/0409787	12/2019	Yip	N/A	G06F 11/1096
2021/0335432	12/2020	Manganelli	N/A	G11C 16/10

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
10-2018-0001711	12/2017	KR	N/A
10-2021-0039174	12/2020	KR	N/A

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) The present application claims priority under 35 U.S.C. § 119(a) to Korean patent application number 10-2022-0066457, filed on May 31, 2022, the entire disclosure of which is incorporated herein by reference.

BACKGROUND

Field of Invention

(2) The present disclosure relates to an electronic device, and more particularly, to a controller controlling a data recovery operation of a semiconductor memory device and a method of operating the controller.

Description of Related Art

(3) A semiconductor memory device may be formed in a two-dimensional structure in which strings are horizontally arranged on a semiconductor substrate, or in a three-dimensional structure in which the strings are vertically stacked on the semiconductor substrate. A three-dimensional semiconductor memory device is a memory device designed in order to resolve a limit of an integration degree of a two-dimensional semiconductor memory device, and may include a plurality of memory cells that are vertically stacked on a semiconductor substrate.

(4) A controller may control an operation of the semiconductor memory device. Specifically, in response to a request received from a host, the controller controls the semiconductor memory device to perform an operation corresponding to the request by transmitting a command to the semiconductor memory device. When an uncorrectable error is included in data read from the semiconductor memory device, the controller may control the semiconductor memory device to perform a recovery operation for the corresponding data.

SUMMARY

- (5) An embodiment of the present disclosure provides a controller capable of efficiently recovering data and a method of operating the same.
- (6) According to an embodiment of the present disclosure, a controller controls a semiconductor memory device including a plurality of pages. The controller includes a command generator configured to generate a command for controlling a program operation or a read operation of a semiconductor memory device, and a data recovery manager configured to generate parity data corresponding to a weak page among a plurality of pages included in the semiconductor memory device and recover data programmed to the weak page.
- (7) In an embodiment of the present disclosure, the data recovery manager may generate the parity data based on data programmed to the weak page and data programmed to a page adjacent to the weak page.
- (8) According to another embodiment of the present disclosure, a semiconductor memory device including a plurality of pages is controlled by a method of operating a controller. The method of operating the controller includes determining whether to program data to a selected page among a plurality of pages included in a semiconductor memory device, determining whether the selected page is a weak page, and generating parity data based on data programmed to the selected page and data programmed to a page adjacent to the selected page, in response to determining that the selected page is the weak page.
- (9) According to still another embodiment of the present disclosure, a semiconductor memory device including a plurality of pages is controlled by a method of operating a controller. The method of operating the controller includes controlling a semiconductor memory device to read data programmed in a weak page among a plurality of pages included in the semiconductor memory device, receiving data of the weak page from the semiconductor memory device and performing an error correction operation on the received data, obtaining parity data associated with the weak page and data programmed to a page adjacent to the weak page, in response to determining that error correction of the received data is failed, and recovering the data of the weak page based on the parity data associated with the weak page and the data programmed to the page adjacent to the weak page.
- (10) According to still another embodiment of the present disclosure, a memory system includes a semiconductor memory device and a controller. The semiconductor memory device includes a plurality of pages. The controller is configured to control the semiconductor memory device. The controller includes a command generator and a data recovery manager. The command generator is configured to generate a command for controlling a program operation or a read operation on the plurality of pages. The data recovery manager is configured to generate parity data corresponding to a weak page among the plurality of pages and recover data programmed to the weak page.
- (11) The present technology may provide a controller capable of efficiently recovering data and a method of operating the same.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a block diagram illustrating a storage device and a host device according to an embodiment of the present disclosure.
- (2) FIG. 2 is a block diagram illustrating a semiconductor memory device of FIG. 1 according to an embodiment of the present disclosure.
- (3) FIG. 3 is a diagram illustrating a memory cell array of FIG. 2 according to an embodiment of the present disclosure.
- (4) FIG. 4 is a circuit diagram illustrating a memory block among memory blocks shown in FIG. 3

according to an embodiment of the present disclosure.

(5) FIG. 5 is a circuit diagram illustrating a memory block among the plurality of memory blocks included in the memory cell array shown in FIG. 2 according to an embodiment of the present disclosure.

(6) FIG. 6 is a block diagram illustrating a controller according to an embodiment of the present disclosure.

(7) FIG. 7 is a block diagram illustrating a data recovery manager shown in FIG. 6 according to an embodiment of the present disclosure.

(8) FIG. 8 is a diagram illustrating a weak page included in the memory block according to an embodiment of the present disclosure.

(9) FIGS. 9A and 9B are diagrams illustrating generating parity data according to embodiments of the present disclosure.

(10) FIG. 10 is a diagram illustrating a memory block that stores the parity data according to an embodiment of the present disclosure.

(11) FIG. 11 is a flowchart illustrating a method of operating a controller according to an embodiment of the present disclosure.

(12) FIGS. 12A and 12B are diagrams illustrating a method of recovering data programmed in a weak page according to an embodiment of the present disclosure.

(13) FIG. 13 is a flowchart illustrating a method of operating a controller according to another embodiment of the present disclosure.

(14) FIG. 14 is a block diagram illustrating a storage device including a semiconductor memory device and a controller according to an embodiment of the present disclosure.

(15) FIG. 15 is a block diagram illustrating an application example of the storage device of FIG. 14 according to an embodiment of the present disclosure.

(16) FIG. 16 is a block diagram illustrating a computing system including the storage device described with reference to FIG. 15 according to an embodiment of the present disclosure.

DETAILED DESCRIPTION

(17) Specific structural or functional descriptions of embodiments according to the concept which are disclosed in the present specification are illustrated only to describe the embodiments according to the concept of the present disclosure, and the embodiments according to the concept of the present disclosure may be implemented in various forms and should not be construed as being limited to the embodiments described in the present specification.

(18) FIG. 1 is a block diagram illustrating a storage device and a host device according to an embodiment of the present disclosure.

(19) Referring to FIG. 1, the storage device **10** includes a semiconductor memory device **100** and a controller **200**. In addition, the storage device **10** communicates with the host device **20**. The controller **200** controls an overall operation of the semiconductor memory device **100**. In addition, the controller **200** controls the operation of the semiconductor memory device **100** based on an operation request RQ received from the host device **20**.

(20) The semiconductor memory device **100** operates in response to control of the controller **200**. The semiconductor memory device **100** includes a memory cell array having a plurality of memory blocks. In an embodiment, the semiconductor memory device **100** may be a flash memory device.

(21) The controller **200** may exchange user data based on the request RQ from the host device **20**. Specifically, the controller **200** may receive a write request, a read request, a trim request, or the like of the host device **20**, and control the semiconductor memory device **100** based on the received requests. More specifically, the controller **200** may generate commands CMD for controlling the operation of the semiconductor memory device **100** and transmit the commands CMD to the semiconductor memory device **100**. Furthermore, the controller **200** may exchange data with the semiconductor memory device **100**.

(22) The semiconductor memory device **100** is configured to receive a command and an address

from the controller **200** and access an area selected by the address in the memory cell array. That is, the semiconductor memory device **100** performs an internal operation corresponding to the command with respect to the area selected by the address.

(23) For example, the semiconductor memory device **100** may perform a program operation, a read operation, and an erase operation. During the program operation, the semiconductor memory device **100** may program data in the area selected by the address. During the read operation, the semiconductor memory device **100** may read data from the area selected by the address. During the erase operation, the semiconductor memory device **100** may erase data stored in the area selected by the address.

(24) FIG. 2 is a block diagram illustrating the semiconductor memory device of FIG. 1 according to an embodiment of the present disclosure.

(25) Referring to FIG. 2, the semiconductor memory device **100** includes a memory cell array **110**, an address decoder **120**, a read and write circuit **130**, a control logic **140**, and a voltage generator **150**.

(26) The memory cell array **110** includes a plurality of memory blocks BLK1 to BLKz. The plurality of memory blocks BLK1 to BLKz are connected to the address decoder **120** through word lines WLs. The plurality of memory blocks BLK1 to BLKz are connected to the read and write circuit **130** through bit lines BL1 to BLm. Each of the plurality of memory blocks BLK1 to BLKz includes a plurality of memory cells. In an embodiment, the plurality of memory cells are non-volatile memory cells, and may be configured of non-volatile memory cells having a vertical channel structure. The memory cell array **110** may be configured as a memory cell array of a two-dimensional structure. According to an embodiment, the memory cell array **110** may be configured as a memory cell array of a three-dimensional structure. Each of the plurality of memory cells included in the memory cell array may store at least one bit of data. In an embodiment, each of the plurality of memory cells included in the memory cell array **110** may be a single-level cell (SLC) storing one bit of data. In another embodiment, each of the plurality of memory cells included in the memory cell array **110** may be a multi-level cell (MLC) storing two bits of data. In still another embodiment, each of the plurality of memory cells included in the memory cell array **110** may be a triple-level cell storing three bits of data. In still another embodiment, each of the plurality of memory cells included in the memory cell array **110** may be a quad-level cell storing four bits of data. According to an embodiment, the memory cell array **110** may include a plurality of memory cells each storing five or more bits of data.

(27) The address decoder **120**, the read and write circuit **130**, the control logic **140**, and the voltage generator **150** operate as a peripheral circuit that drives the memory cell array **110**. The address decoder **120** is connected to the memory cell array **110** through the word lines WLs. The address decoder **120** is configured to operate in response to control of the control logic **140**. The address decoder **120** receives an address through an input/output buffer (not shown) inside the semiconductor memory device **100**.

(28) The address decoder **120** is configured to decode a block address among received addresses. The address decoder **120** selects at least one memory block according to the decoded block address. In addition, the address decoder **120** applies a read voltage Vread generated by the voltage generator **150** to a selected word line among the selected memory block at a read voltage application operation during a read operation, and applies a pass voltage Vpass to the remaining unselected word lines. In addition, the address decoder **120** applies a verify voltage generated by the voltage generator **150** to the selected word line among the selected memory block and applies the pass voltage Vpass to the remaining unselected word lines during a program verify operation.

(29) The address decoder **120** is configured to decode a column address of the received addresses. The address decoder **120** transmits the decoded column address to the read and write circuit **130**.

(30) The read operation and a program operation of the semiconductor memory device **100** are performed in a page unit. Addresses received at a time of a request of the read operation and the

program operation include a block address, a row address, and a column address. The address decoder **120** selects one memory block and one word line according to the block address and the row address. The column address is decoded by the address decoder **120** and is provided to the read and write circuit **130**.

(31) The address decoder **120** may include a block decoder, a row decoder, a column decoder, an address buffer, and the like.

(32) The read and write circuit **130** includes a plurality of page buffers PB1 to PBm. The read and write circuit **130** may operate as a “read circuit” during a read operation of the memory cell array **110** and may operate as a “write circuit” during a write operation of the memory cell array **110**. The plurality of page buffers PB1 to PBm are connected to the memory cell array **110** through the bit lines BL1 to BLm. During the read operation and the program verify operation, in order to sense a threshold voltage of the memory cells, the plurality of page buffers PB1 to PBm sense a change of an amount of a current flowing according to a program state of a corresponding memory cell through a sensing node while continuously supplying a sensing current to the bit lines connected to the memory cells, and latches the sensed change as sensing data. The read and write circuit **130** operates in response to page buffer control signals output from the control logic **140**.

(33) During the read operation, the read and write circuit **130** senses data of the memory cell, temporarily stores read data, and outputs data DATA to the input/output buffer (not shown) of the semiconductor memory device **100**. In an embodiment, the read and write circuit **130** may include a column selection circuit, and the like, in addition to the page buffers (or page registers).

(34) The control logic **140** is connected to the address decoder **120**, the read and write circuit **130**, and the voltage generator **150**. The control logic **140** receives a command CMD and a control signal CTRL through the input/output buffer (not shown) of the semiconductor memory device **100**. The control logic **140** is configured to control overall operations of the semiconductor memory device **100** in response to the control signal CTRL. In addition, the control logic **140** outputs a control signal for adjusting a sensing node pre-charge potential level of the plurality of page buffers PB1 to PBm. The control logic **140** may control the read and write circuit **130** to perform the read operation of the memory cell array **110**.

(35) The voltage generator **150** generates the read voltage Vread and the pass voltage Vpass during the read operation in response to the control signal output from the control logic **140**. In order to generate a plurality of voltages having various voltage levels, the voltage generator **150** may include a plurality of pumping capacitors that receive an internal power voltage, and generate the plurality of voltages by selectively activating the plurality of pumping capacitors in response to the control of the control logic **140**.

(36) The address decoder **120**, the read and write circuit **130**, and the voltage generator **150** may function as a “peripheral circuit” that performs the read operation, the write operation, and the erase operation on the memory cell array **110**. The peripheral circuit performs the read operation, the write operation, and the erase operation on the memory cell array **110** based on the control of the control logic **140**.

(37) FIG. 3 is a diagram illustrating the memory cell array of FIG. 2 according to an embodiment of the present disclosure.

(38) Referring to FIG. 3, the memory cell array **110** includes a plurality of memory blocks BLK1 to BLKz. Each memory block may have a three-dimensional structure. Each memory block includes a plurality of memory cells stacked on a substrate. The plurality of memory cells are arranged along a +X direction, a +Y direction, and a +Z direction. A structure of each memory block of the three-dimensional structure is described in more detail with reference to FIG. 4.

(39) FIG. 4 is a circuit diagram illustrating a memory block BLKa among the memory blocks BLK1 to BLKz of FIG. 3 according to an embodiment of the present disclosure.

(40) Referring to FIG. 4, the memory block BLKa includes a plurality of cell strings CS11 to CS1m and CS21 to CS2m. In the memory block BLKa, m cell strings are arranged in a row direction (that

is, the +X direction). In FIG. 4, two cell strings are arranged in a column direction (that is, the +Y direction). However, this is for convenience of description and it may be understood that three or more cell strings may be arranged in the column direction.

(41) Each of the plurality of cell strings CS11 to CS1 m and CS21 to CS2 m includes at least one source select transistor SST, first to n -th memory cells MC1 to MC n , and at least one drain select transistor DST.

(42) Each of the select transistors SST and DST and the memory cells MC1 to MC n may have a similar structure. In an embodiment, each of the select transistors SST and DST and the memory cells MC1 to MC n may include a channel layer, a tunneling insulating layer, a charge storage layer, and a blocking insulating layer. In an embodiment, a pillar for providing the channel layer may be provided in each cell string. In an embodiment, a pillar for providing at least one of the channel layer, the tunneling insulating layer, the charge storage layer, and the blocking insulating layer may be provided in each cell string.

(43) The source select transistor SST of each cell string is connected between a common source line CSL and the memory cells MC1 to MC n .

(44) In an embodiment, the source select transistors of the cell strings arranged in the same row are connected to a source select line extending in the row direction, and the source select transistors of the cell strings arranged in different rows are connected to different source select lines. In FIG. 4, the source select transistors of the cell strings CS11 to CS1 m of a first row are connected to a first source select line SSL1. The source select transistors of the cell strings CS21 to CS2 m of a second row are connected to a second source select line SSL2.

(45) In another embodiment, the source select transistors of the cell strings CS11 to CS1 m and CS21 to CS2 m may be commonly connected to one source select line.

(46) The first to n -th memory cells MC1 to MC n of each cell string are connected in series between the source select transistor SST and the drain select transistor DST. Gates of the first to n -th memory cells MC1 to MC n are connected to first to n -th word lines WL1 to WL n , respectively.

(47) The drain select transistor DST of each cell string is connected between a corresponding bit line and the memory cells MC1 to MC n . Cell strings arranged in the row direction are connected to the drain select line extending in the row direction. The drain select transistors of the cell strings CS11 to CS1 m of the first row are connected to a first drain select line DSL1. The drain select transistors of the cell strings CS21 to CS2 m of the second row are connected to a second drain select line DSL2.

(48) The cell strings arranged in the column direction are connected to the bit lines extending in the column direction. In FIG. 4, the cell strings CS11 and CS21 of the first column are connected to the first bit line BL1. The cell strings CS1 m and CS2 m of the m -th column are connected to the m -th bit line BL m .

(49) The memory cells connected to the same word line in the cell strings arranged in the row direction configure one page. For example, the memory cells connected to the first word line WL1, among the cell strings CS11 to CS1 m of the first row configure one page. The memory cells connected to the first word line WL1, among the cell strings CS21 to CS2 m of the second row configure another page. The cell strings arranged in one row direction may be selected by selecting one of the drain select lines DSL1 and DSL2. One page of the selected cell strings may be selected by selecting one of the word lines WL1 to WL n .

(50) In another embodiment, even bit lines and odd bit lines may be provided instead of the first to m -th bit lines BL1 to BL m . In addition, even-numbered cell strings among the cell strings CS11 to CS1 m or CS21 to CS2 m arranged in the row direction may be connected to the bit lines, and odd-numbered cell strings among the cell strings CS11 to CS1 m or CS21 to CS2 m arranged in the row direction may be connected to odd bit lines, respectively.

(51) In an embodiment, at least one of the first to n -th memory cells MC1 to MC n may be used as a dummy memory cell. For example, at least one or more dummy memory cells are provided to

reduce an electric field between the source select transistor SST and the memory cells MC1 to MCn. Alternatively, at least one or more dummy memory cells are provided to reduce an electric field between the drain select transistor DST and the memory cells MC1 to MCn. As more dummy memory cells are provided, reliability of an operation for the memory block BLKa is improved, however, the size of the memory block BLKa increases. As less memory cells are provided, the size of the memory block BLKa may be reduced, however, the reliability of the operation for the memory block BLKa may be reduced.

(52) In order to efficiently control at least one dummy memory cell, each of the dummy memory cells may have a required threshold voltage. Before or after an erase operation for the memory block BLKa, program operations for all or a part of the dummy memory cells may be performed. When the erase operation is performed after the program operation is performed, the dummy memory cells may have a required threshold voltage by controlling a voltage applied to dummy word lines connected to the respective dummy memory cells.

(53) FIG. 5 is a circuit diagram illustrating a memory block BLKb among the plurality of memory blocks BLK1 to BLKz included in the memory cell array **110** of FIG. 2 according to an embodiment of the present disclosure.

(54) Referring to FIG. 5, the memory block BLKb includes a plurality of cell strings CS1 to CSm. The plurality of cell strings CS1 to CSm may be connected to a plurality of bit lines BL1 to BLm, respectively. Each of the cell strings CS1 to CSm includes at least one source select transistor SST, first to n-th memory cells MC1 to MCn, and at least one drain select transistor DST.

(55) Each of the select transistors SST and DST and the memory cells MC1 to MCn may have a similar structure. In an embodiment, each of the select transistors SST and DST and the memory cells MC1 to MCn may include a channel layer, a tunneling insulating layer, a charge storage layer, and a blocking insulating layer. The source select transistor SST of each cell string is connected between a common source line CSL and the memory cells MC1 to MCn.

(56) The first to n-th memory cells MC1 to MCn of each cell string are connected between the source select transistor SST and the drain select transistor DST.

(57) The drain select transistor DST of each cell string is connected between a corresponding bit line and the memory cells MC1 to MCn.

(58) Memory cells connected to the same word line configure one page. The cell strings CS1 to CSm may be selected by selecting the drain select line DSL. One page among the selected cell strings may be selected by selecting one of the word lines WL1 to WLn.

(59) In another embodiment, even bit lines and odd bit lines may be provided instead of the first to m-th bit lines BL1 to BLm. Even-numbered cell strings among the cell strings CS1 to CSm may be connected to even bit lines, and odd-numbered cell strings may be connected to odd bit lines, respectively.

(60) As shown in FIGS. 3 and 4, the memory blocks BLK1 to BLKz of the semiconductor memory device **100** may be configured as memory blocks of a three-dimensional structure. However, the present disclosure is not limited thereto, and as shown in FIG. 5, the memory blocks BLK1 to BLKz of the semiconductor memory device **100** may be configured as memory blocks of a two-dimensional structure.

(61) In a production process of the semiconductor memory device **100**, a weak page may occur. For some process reasons, a threshold voltage characteristic of memory cells included in a specific page may be relatively bad. A page including these memory cells is the weak page, and stored data is highly likely to be damaged compared to other pages. That is, the data stored in the weak page may include error bits of a level, which may not be recovered even by an error correction operation when reading the data later. In general, when an uncorrectable error occurs in the weak page, a read operation on a corresponding page is repeatedly performed while changing a read voltage. When data read is successful by repeatedly performing the read operation, a read-reclaim operation of programming the data programmed in the weak page to a different physical position may be

performed. After the read-reclaim operation is performed, the memory block including the weak page is set as a bad block and the corresponding memory block is no longer used.

(62) In a case according to a processing method described above, two problems occur. First, in a case where the semiconductor memory device **100** includes a large number of weak pages, read performance deterioration of the storage device may occur as the read operation is repeatedly performed when the read operation on the weak pages is failed. Second, in a case where the semiconductor memory device **100** includes a large number of weak pages, when an uncorrectable error occurs, since a corresponding memory block is set as a bad block, the number of usable memory blocks is rapidly reduced. Therefore, as a use period of the storage device **10** increases, a usable memory area is rapidly reduced. In particular, when the storage device **10** is used in an electric system of a vehicle, a relatively high durability life is required. This is because replacing the storage device **10** is difficult due to a characteristic of the electric system and a use life is long. As a result, a method of operating a controller capable of quickly recovering data when an error occurs in a weak page and maximally securing a usable memory block is required.

(63) In accordance with a controller and a method of operating the same according to an embodiment of the present disclosure, when data is programmed to a weak page, parity data is generated based on data programmed in the weak page and data programmed in one or more pages adjacent to the weak page. When an uncorrectable error occurs in the weak page during a read operation, the data programmed in the weak page is recovered based on the parity data. Accordingly, even though an error occurs in the weak page, the corresponding data may be quickly recovered. In addition, even though an error occurs in the weak page, since a memory block including the weak page may be used as it is, a usable memory block in the storage device may be maximally secured.

(64) In the above description, the weak page is a page including a defect occurring in the production process of the storage device, but the present disclosure is not limited thereto. For example, when an error bit is detected a certain level or more in data stored in a normal page during a use process of the storage device, the corresponding page may be determined as the weak page.

(65) FIG. **6** is a block diagram illustrating the controller **200** according to an embodiment of the present disclosure. Referring to FIG. **6**, the controller **200** according to an embodiment of the present disclosure may include a command generator **210** and a data recovery manager **230**.

(66) The command generator **210** may receive a write request RQ.sub.WRT and write data DATA.sub.WRT from the host device **20**. The command generator **210** may generate a program command CMD.sub.PGM in response to the write request RQ.sub.WRT received from the host device **20**. The command generator **210** may transmit the program command CMD.sub.PGM and program data DATA.sub.PGM to the semiconductor memory device **100**. The program data DATA.sub.PGM may be data generated from the write data DATA.sub.WRT received from the host device **20**.

(67) The command generator **210** may transmit a program address ADD.sub.PGM indicating a physical position to which the program data DATA.sub.PGM is to be programmed to the data recovery manager **230**. Moreover, although not shown in FIG. **6**, the program address ADD.sub.PGM may be transmitted to the semiconductor memory device **100** together with the program command CMD.sub.PGM and the program data DATA.sub.PGM. The semiconductor memory device **100** may program the program data DATA.sub.PGM to the physical position corresponding to the program address ADD.sub.PGM in response to the program command CMD.sub.PGM. The data recovery manager **230** receives the program address ADD.sub.PGM from the command generator **210** and determines whether a page corresponding to the program address ADD.sub.PGM is the weak page. When the page corresponding to the program address ADD.sub.PGM is the weak page, the data recovery manager **230** generates the parity data based on the program data DATA.sub.PGM and the data programmed in the page adjacent to the weak page. To this end, the program data DATA.sub.PGM may be transmitted from the command generator

210 to the data recovery manager **230**.

(68) In an embodiment, in order to obtain the data programmed in the page adjacent to the weak page, the data recovery manager **230** may transmit a control signal SIG.sub.CTR to the command generator **210**. The command generator **210** may generate a read command for reading the data from the page adjacent to the weak page based on the control signal SIG.sub.CTR. The command generator **210** may transmit the generated read command to the semiconductor memory device **100**. The semiconductor memory device **100** may perform a read operation in response to the received read command. The data programmed in the page adjacent to the weak page, that is, adjacent data DATA.sub.N, is read by the read operation of the semiconductor memory device **100**. The semiconductor memory device **100** transmits the read adjacent data DATA.sub.N to the controller **200**. The data recovery manager **230** of the controller **200** receives the adjacent data DATA.sub.N and generates the parity data DATA.sub.PRT based on the adjacent data DATA.sub.N and the program data DATA.sub.PGM.

(69) In an embodiment, the data recovery manager **230** may generate the parity data DATA.sub.PRT by performing an exclusive-OR (XOR) operation on the adjacent data DATA.sub.N and the program data DATA.sub.PGM. Specifically, the data recovery manager **230** performs an exclusive-OR (XOR) operation on bits included in the adjacent data DATA.sub.N and bits included in the program data DATA.sub.PGM, respectively, and generates the parity data DATA.sub.PRT including bits generated as a result of performing the exclusive-OR (XOR) operation. For example, the data recovery manager **230** may perform an exclusive-OR (XOR) operation on a first bit included in the adjacent data DATA.sub.N and a first bit included in the program data DATA.sub.PGM, and determine a bit calculated as a result of performing the exclusive-OR (XOR) operation as a first bit of the parity data DATA.sub.PRT. In addition, the data recovery manager **230** may perform an exclusive-OR (XOR) operation on a second bit included in the adjacent data DATA.sub.N and a second bit included in the program data DATA.sub.PGM, and determine a bit calculated as a result of performing the exclusive-OR (XOR) operation as a second bit of the parity data DATA.sub.PRT. In such a method, the bits included in the parity data DATA.sub.PRT may be calculated, by performing the exclusive-OR (XOR) operation on the respective bits included in the adjacent data DATA.sub.N and the corresponding bits of the program data DATA.sub.PGM.

(70) The data recovery manager **230** may maintain the generated parity data DATA.sub.PRT. In an embodiment, the data recovery manager **230** may transmit the generated parity data DATA.sub.PRT to the semiconductor memory device **100**. In this case, the parity data DATA.sub.PRT may be programmed to the memory cell array of the semiconductor memory device **100**. To this end, the command generator **210** of the controller **200** may generate a program command for programming the parity data DATA.sub.PRT, and transmit the generated program command to the semiconductor memory device **100** together with the parity data DATA.sub.PRT.

(71) In an embodiment, the parity data DATA.sub.PRT may be programmed to a memory block different from the memory block including the weak page. That is, the page to which the parity data DATA.sub.PRT is programmed and the weak page belong to different memory blocks. According to an embodiment, the semiconductor memory device **100** may include a dedicated memory block for programming only the parity data DATA.sub.PRT.

(72) For stability of the parity data DATA.sub.PRT, the dedicated memory block to which the parity data DATA.sub.PRT is programmed may include a single-level cell (SLC). That is, the parity data DATA.sub.PRT may be programmed in an SLC method.

(73) FIG. 7 is a block diagram illustrating the data recovery manager **230** of FIG. 6 according to an embodiment of the present disclosure. Referring to FIG. 7, the data recovery manager **230** may include a weak page information storage **231**, a parity data generator **233**, and a parity data storage **235**.

(74) The weak page information storage **231** stores information for identifying weak pages among pages included in the semiconductor memory device **100**. That is, weak page information

indicating which page among the pages included in the semiconductor memory device **100** is the weak page may be stored in the weak page information storage **231**.

(75) In an embodiment, the weak page information may be obtained in a manufacturing step of the semiconductor memory device **100**. That is, the weak page may be detected in a manufacturing test step of the semiconductor memory device **100**. In another embodiment, the weak page information may be obtained while the semiconductor memory device **100** is used. For example, when error bits equal to or greater than a specific threshold value are included in data read from a normal page other than the weak page, the controller **200** may determine the corresponding page as a new weak page. In this case, information on the new weak page may be additionally stored in the weak page information storage **231**.

(76) When the weak page information storage **231** is configured as a non-volatile memory, the weak page information storage **231** may maintain the weak page information even though the storage device **10** is turned off. On the other hand, when the weak page information storage **231** is configured as a volatile memory, the weak page information storage **231** may not maintain the weak page information when the storage device **10** is turned off. In this case, the weak page information may be stored in a specific area in the memory cell array of the semiconductor memory device **100**. When the storage device **10** is turned on, the semiconductor memory device **100** may read the weak page information from the memory cell array and transmit the weak page information to the controller **200**. The weak page information storage **231** may store the weak page information received from the semiconductor memory device **100**.

(77) The weak page information storage **231** receives the program address ADD.sub.PGM and determines whether the page corresponding to the program address ADD.sub.PGM is the weak page. Specifically, when the weak page information related to the page corresponding to the program address ADD.sub.PGM is stored, the page corresponding to the program address ADD.sub.PGM is determined as the weak page. On the other hand, when the weak page information related to the page corresponding to the program address ADD.sub.PGM is not stored, it is determined that the page corresponding to the program address ADD.sub.PGM is not the weak page.

(78) When the page corresponding to the program address ADD.sub.PGM is the weak page, the weak page information storage **231** transmits the control signal SIG.sub.CTR to the command generator **210** and the parity data generator **233**. As described above, the command generator **210** may generate the read command for reading the data from the page adjacent to the weak page based on the control signal SIG.sub.CTR. The adjacent data DATA.sub.N read in response to the corresponding read command may be transmitted to the parity data generator **233**.

(79) The parity data generator **233** receives the program data DATA.sub.PGM. When the page corresponding to the program address ADD.sub.PGM is the weak page, the parity data generator **233** may receive the control signal SIG.sub.CTR and the adjacent data DATA.sub.N. In response to the control signal SIG.sub.CTR, the parity data generator **233** generates the parity data DATA.sub.PRT based on the adjacent data DATA.sub.N and the program data DATA.sub.PGM. In an embodiment, the parity data generator **233** may generate the parity data DATA.sub.PRT by performing an exclusive-OR (XOR) operation on the adjacent data DATA.sub.N and the program data DATA.sub.PGM.

(80) The generated parity data DATA.sub.PRT may be transmitted to the parity data storage **235**. The parity data storage **235** stores the received parity data DATA.sub.PRT. When the parity data storage **235** is configured as a non-volatile memory, the parity data storage **235** may maintain the parity data DATA.sub.PRT even though the storage device **10** is turned off. On the other hand, when the parity data storage **235** is configured as a volatile memory, the parity data storage **235** may not maintain the parity data DATA.sub.PRT when the storage device **10** is turned off. In this case, the parity data DATA.sub.PRT may be stored in a specific area in the memory cell array of the semiconductor memory device **100**.

(81) FIG. 8 is a diagram illustrating the weak page included in the memory block. Referring to FIG. 8, a memory block BLKp includes 16 physical pages PAGE 1 to PAGE 16. However, this is an example, and the number of physical pages included in one memory block may be greater than this number. However, for convenience of discussion, a memory block including 16 physical pages is described as an example.

(82) Referring to FIG. 8, the first to sixteenth physical pages PAGE 1 to PAGE 16 are respectively connected to first to sixteenth word lines WL1 to WL16. In FIG. 8, a third page PAGE 3, a ninth page PAGE 9, and a thirteenth page PAGE 13 are weak pages. As described above, the weak page is a page including memory cells having a deteriorated characteristic, and is a page with a high possibility that a large number of error bits occur when reading data after programming the data.

(83) The weak page information storage 231 may store weak page information indicating that the third page PAGE 3, the ninth page PAGE 9, and the thirteenth page PAGE 13 of the memory block BLKp are the weak pages. As described above, the weak page information may be generated based on a test in the production process of the semiconductor memory device 100. As another example, the weak page information may be generated by detecting occurrence of an error bit during actual use of the semiconductor memory device 100.

(84) According to an embodiment of the present disclosure, when data is programmed to the weak page of the memory block BLKp, the parity data is generated based on the corresponding program data DATA.sub.PGM and the data programmed in the adjacent page. When an uncorrectable error occurs in the weak page during the read operation, the data programmed in the weak page is recovered based on the parity data. Accordingly, even though an error occurs in the weak page, the data may be quickly recovered. In addition, even though an error occurs in the weak page, since the memory block including the weak page may be used as it is, a usable memory block in the storage device may be maximally secured.

(85) FIGS. 9A and 9B are diagrams illustrating generating the parity data according to embodiments of the present disclosure.

(86) Referring to FIG. 9A, a first embodiment of generating the parity data based on the data programmed in the weak page and the data programmed in the adjacent page is shown. In FIG. 9A, when data is programmed to the weak page, the parity data may be generated by performing an exclusive-OR (XOR) operation on the data programmed in the corresponding weak page and one page adjacent thereto.

(87) Specifically, when data is programmed to the third page PAGE 3 which is the weak page, parity data DATA.sub.PRT1 may be generated by performing an XOR operation on the data programmed in the third page PAGE 3 and data programmed in a second page PAGE 2 adjacent thereto. In addition, when data is programmed to the ninth page PAGE 9 which is the weak page, parity data DATA.sub.PRT2 may be generated by performing an XOR operation on the data programmed in the ninth page PAGE 9 and data programmed in an eighth page PAGE 8 adjacent thereto. When data is programmed to the thirteenth page PAGE 13 which is the weak page, the parity data DATA.sub.PRT3 may be generated by performing an XOR operation on the data programmed in the thirteenth page PAGE 13 and data programmed in a twelfth page PAGE 12 adjacent thereto.

(88) However, this is an example, and the parity data may be generated by another method. For example, the parity data may be generated by performing an XOR operation on the data programmed in the third page PAGE 3 and data programmed in a fourth page PAGE 4 adjacent thereto, the parity data may be generated by performing an XOR operation on the data programmed in the ninth page PAGE 9 and data programmed in a tenth page PAGE 10 adjacent thereto, and the parity data may be generated by performing an XOR operation on the data programmed in the thirteenth page PAGE 13 and data programmed in a fourteenth page PAGE 14 adjacent thereto.

(89) A situation in which a program operation of data is performed in a direction from the first page PAGE 1 to the sixteenth page PAGE 16 is illustrated. In this case, when data is to be programmed

to the third page PAGE 3, data may be already programmed in the second page PAGE 2. Therefore, the parity data DATA.sub.PRT1 may be generated by reading the data from the second page PAGE 2, and then performing an XOR operation on the read data and the data read from the third page PAGE 3.

(90) Referring to FIG. 9B, a second embodiment of generating the parity data based on the data programmed in the weak page and the data programmed in the adjacent pages is shown. In FIG. 9B, when data is programmed to the weak page, the parity data may be generated by performing an exclusive-OR (XOR) operation on the data programmed in the corresponding weak page and data programmed in two adjacent pages.

(91) Specifically, when data is programmed to the third page PAGE 3 which is the weak page, the parity data DATA.sub.PRT1 may be generated by performing an XOR operation on the data programmed in the third page PAGE 3 and data programmed in the second page PAGE 2 and the fourth page PAGE 4 adjacent thereto. In this case, an exclusive-OR (XOR) operation may be performed on a first bit included in the data programmed in the second page PAGE 2, a first bit included in the data programmed in the third page PAGE 3, and a first bit included in the data programmed in the fourth page PAGE 4, and a bit calculated as a result of the exclusive-OR (XOR) operation may be determined as a first bit of the parity data DATA.sub.PRT1. In addition, an exclusive-OR (XOR) operation may be performed on a second bit included in the data programmed in the second page PAGE 2, a second bit included in the data programmed in the third page PAGE 3, and a second bit included in the data programmed in the fourth page PAGE 4, and a bit calculated as a result of the exclusive-OR (XOR) operation may be determined as a second bit of the parity data DATA.sub.PRT1. In such a method, the parity data DATA.sub.PRT1 may be generated based on the data programmed in the third page PAGE 3 which is the weak page and the data programmed in the second and fourth pages PAGE 2 and PAGE 4 adjacent to the third page.

(92) Similarly, when data is programmed to the ninth page PAGE 9 which is the weak page, the parity data DATA.sub.PRT2 may be generated by performing an XOR operation on the data programmed in the ninth page PAGE 9 and the data programmed in the eighth page PAGE 8 and the tenth page PAGE 10 adjacent thereto. When data is programmed to the thirteenth page PAGE 13 which is the weak page, the parity data DATA.sub.PRT3 may be generated by performing an XOR operation on the data programmed in the thirteenth page PAGE 13 and the data programmed in the twelfth page PAGE 12 and the fourteenth page PAGE 14 adjacent thereto.

(93) A situation in which a program operation of data is performed in a direction from the first page PAGE 1 to the sixteenth page PAGE 16 is illustrated. In this case, when data is to be programmed to the third page PAGE 3, the parity data DATA.sub.PRT1 may be generated after both of the data programmed in the second page PAGE 2 and the data programmed in the fourth page PAGE 4 are obtained. In an embodiment, the controller 200 may generate the parity data DATA.sub.PRT1, DATA.sub.PRT2, and DATA.sub.PRT3 after all data up to the sixteenth page of the memory block BLKp are programmed, that is, after the memory block BLKp is closed.

(94) FIG. 10 is a diagram illustrating a memory block that stores the parity data according to an embodiment of the present disclosure.

(95) Referring to FIG. 10, the generated parity data DATA.sub.PRT1, DATA.sub.PRT2, and DATA.sub.PRT3 may be programmed to a memory block BLKq different from the memory block BLKp. Specifically, the parity data DATA.sub.PRT1 may be programmed to the first page PAGE 1 of the memory block BLKq, the parity data DATA.sub.PRT2 may be programmed to the second page PAGE 2 of the memory block BLKq, and the parity data DATA.sub.PRT3 may be programmed to the third page PAGE 3 of the memory block BLKq. In an embodiment, the memory block BLKq may be a dedicated memory block for storing only the parity data. In addition, according to an embodiment, the parity data DATA.sub.PRT1, DATA.sub.PRT2, and DATA.sub.PRT3 may be programmed to the memory block BLKq in an SLC method.

(96) FIG. 11 is a flowchart illustrating a method of operating a controller according to an

embodiment of the present disclosure. More specifically, FIG. 11 is a flowchart illustrating an operation method when data is programmed to the weak page.

(97) Referring to FIG. 11, the method of operating the controller according to an embodiment of the present disclosure includes determining to program data in the selected page of the semiconductor memory device (S110), referring to the weak page information storage (S120), and determining whether the page to which data is to be programmed is the weak page (S130).

(98) In operation S110, the controller 200 may determine to program data in the selected page of the semiconductor memory device based on a write request RQWT from the host device 20. As another example, the controller 200 may determine to program data in the selected page of the semiconductor memory device regardless of the write request RQWT from the host device 20. For example, the controller 200 independently may determine to program valid data from a victim block to the selected page of the semiconductor memory device according to a garbage collection operation.

(99) In operation S120, in order to determine whether the page to which the data is to be programmed is the weak page, the controller 200 refers to the weak page information storage 231 in the data recovery manager 230. When the page to which the data is to be programmed is not the weak page (S130: No), the parity data may not be generated.

(100) In addition, the method of operating the controller according to an embodiment of the present disclosure may further include generating the parity data based on the data programmed in the weak page and the data programmed in the adjacent page (S140), when the page to which the data is to be programmed is the weak page (S130: Yes), storing the parity data in the controller (S150), and controlling the semiconductor memory device to program the parity data (S160).

(101) In operation S140, the parity data may be generated in a method shown in FIG. 9A or the parity data may be generated in a method shown in FIG. 9B. In operation S140, the read command for reading corresponding data may be generated to obtain the data programmed in the adjacent page. When the data programmed in the adjacent page is also cached in a buffer in the controller 200, the parity data may be generated based on the cached data without generating the read command.

(102) The parity data generated in operation S150 may be stored in the parity data storage 235 in the data recovery manager 230. In addition, in operation S160, the program command for programming the parity data may be generated, and the generated program command may be transmitted to the semiconductor memory device together with the parity data. According to an embodiment of the present disclosure, both of operations S150 and S160 may be performed. In this case, the parity data may be stored in both of the parity data storage 235 and the memory block BLKq of the semiconductor memory device. According to an embodiment, only one of the operation S150 and S160 may be selectively performed.

(103) Although not shown in FIG. 11, after operation S110, generating the program command for programming data in the selected page, and transmitting the generated program command to the semiconductor memory device may be performed. However, this operation may be performed at any operation after operation S110. For example, immediately after operation S110 is performed and before operation S120 is performed, the controller may generate the program command and transmit the program command to the semiconductor memory device together with program data. Alternatively, after the operation S160 is performed, the controller may generate the program command and transmit the program command to the semiconductor memory device together with the program data. That is, a performance time of generating the program command for programming the data in the selected page and transmitting the generated program command to the semiconductor memory device may be variously changed according to a design.

(104) FIGS. 12A and 12B are diagrams illustrating a method of recovering data programmed in a weak page according to an embodiment of the present disclosure. Hereinafter, the present disclosure is described based on an embodiment in which the parity data is generated by the

method shown in FIG. 9B.

(105) Referring to FIG. 12A, a situation in which the data programmed in the ninth page PAGE 9 which is the weak page of the memory block BLKp is read is shown. When an uncorrectable error is included in read data as a result of reading the data from the ninth page PAGE 9, it may be determined that a read failure occurs. According to a method of operating a controller according to an embodiment of the present disclosure, when an operation of reading the data stored in the weak page is failed, the data programmed in the weak page is recovered based on the data stored in the adjacent page and the parity data. As shown in FIG. 12A, the controller 200 may control the semiconductor memory device 100 to read the data from the eighth page PAGE 8 and the tenth page PAGE 10 adjacent to the ninth page PAGE 9.

(106) In addition, as shown in FIG. 12B, the controller 200 may control the semiconductor memory device 100 to read the parity data DATA.sub.PRT2 from the memory block BLKq. The parity data DATA.sub.PRT2 is the parity data corresponding to the ninth page PAGE 9 of the memory block BLKp.

(107) The controller 200 may recover the data programmed in the ninth page PAGE 9 based on the data programmed in the eighth page PAGE 8 in the read memory block BLKp, the data programmed in the tenth page PAGE 10, and the parity data DATA.sub.PRT2 in the memory block BLKq. Specifically, the data programmed in the ninth page PAGE 9 may be recovered by performing an XOR operation on the parity data DATA.sub.PRT2, and the data programmed in the eighth page PAGE 8 and the data programmed in the tenth page PAGE 10 adjacent to the ninth page PAGE 9.

(108) The data recovery operation described with reference to FIGS. 12A and 12B may be performed by the command generator 210 and the data recovery manager 230 shown in FIG. 6. First, in order to check whether the data on which the read operation is failed is the data programmed in the weak page, the data recovery manager 230 may refer to the weak page information storage 231 and check whether the page corresponding to the read address is the weak page. When it is checked that the data on which the read operation is failed is the data programmed in the weak page, the weak page information storage 231 generates the control signal for controlling the command generator 210 to generate the read command for reading each of the data from the eighth page PAGE 8 in the memory block BLKp, the data from the tenth page PAGE 10, and the parity data DATA.sub.PRT2 from the memory block BLKq. The command generator 210 may generate the read command for reading the data from the eighth page PAGE 8 in the memory block BLKp, the data from the tenth page PAGE 10, and the parity data DATA.sub.PRT2 from the memory block BLKq under control of the weak page information storage 231, and transmit the generated read command to the semiconductor memory device 100.

(109) The parity data generator 233 may receive the data programmed in the eighth page PAGE 8 in the memory block BLKp and the data programmed in the tenth page PAGE 10 from the semiconductor memory device 100. The parity data generator 233 may receive the parity data DATA.sub.PRT2 from the semiconductor memory device 100 or may receive the parity data DATA.sub.PRT2 from the parity data storage 235 according to an embodiment. The parity data generator 233 may recover the data programmed in the ninth page PAGE 9 by performing an XOR operation on the data programmed in the eighth page PAGE 8, the data programmed in the tenth page PAGE 10, and the parity data DATA.sub.PRT2.

(110) FIG. 13 is a flowchart illustrating a method of operating a controller according to another embodiment of the present disclosure. Hereinafter, the present disclosure is described together with the example described with reference to FIGS. 12A and 12B.

(111) Referring to FIG. 13, the method of operating the controller according to the other embodiment of the present disclosure includes controlling the semiconductor memory device to read the data from the selected page (S210), performing an error correction operation on the read data (S220), determining whether error correction is failed (S230), referring to the weak page

information storage (S240), determining whether the selected page is the weak page (S250), obtaining the parity data associated with the selected page and obtaining the data programmed in the page adjacent to the selected page (S260), recovering the data programmed in the selected page based on the parity data and the data programmed in the adjacent page (S270), determining whether the recovery of the data programmed in the selected page is successful (S280), and controlling the semiconductor memory device to read the data from the selected page again (S290).

(112) In operation S210, the controller 200 controls the semiconductor memory device to read the data from the selected page among the pages included in the semiconductor memory device 100. Specifically, in operation S210, the controller may generate the read command for reading the data from the selected page, and transmit the generated read command to the semiconductor memory device 100. In response to the read command, the semiconductor memory device 100 may read the data from the selected page, and transmit the read data to the controller 200.

(113) In operation S220, the controller 200 performs an error correction operation on the read data. When error correction is successful as a result of performing the error correction operation (S230: No), the read operation may be ended.

(114) When an uncorrectable error is included in the read data includes as a result of performing the error correction operation (S230: Yes), the controller 200 refers to the weak page information storage 231 to determine whether the corresponding data is the data programmed in the weak page.

(115) When the page storing the data in which the uncorrectable error occurs is not the weak page (S250: No), since the corresponding parity data does not exist, a data recovery operation based on the parity data is not performed and the method proceeds to operation S290. In operation S290, the semiconductor memory device may be controlled to re-read the data programmed in the selected page by various methods. For example, in operation S290, the controller 200 may control the semiconductor memory device to correct the read voltage based on a read-retry table, and then read the data from the selected page based on the corrected read voltage. Alternatively, in operation S290, the controller 200 may control the semiconductor memory device to correct the read voltage by searching for a valley between threshold voltage distributions of the memory cells included in the selected page, and read the data from the selected page based on the corrected read voltage. However, this is an example, and in operation S290, the semiconductor memory device may be controlled to re-perform the read operation according to various other methods.

(116) When the page storing the data in which the uncorrectable error occurs is the weak page (S250: Yes), the controller 200 may obtain the parity data associated with the selected page and the data programmed in the page adjacent to the selected page (S260). Specifically, as described with reference to FIGS. 12A and 12B, in operation S260, the controller 200 may control the semiconductor memory device to read the data from the pages PAGE 8 and PAGE 10 adjacent to the selected page PAGE 9, and the parity data DATA.sub.PRT2.

(117) In operation S270, the controller 200 may recover the data programmed in the selected page based on the obtained parity data and the data programmed in the adjacent page. Specifically, the parity data generator 233 may perform an XOR operation on the data programmed in the eighth page PAGE 8 adjacent to the ninth page PAGE 9 which is the weak page in which an uncorrectable error occurs, the data programmed in the tenth page PAGE 10, and the parity data DATA.sub.PRT2. As a result of the XOR operation, the data programmed in the ninth page PAGE 9 may be recovered. In this disclosure, when data programmed in a page is recovered during a read operation, the data read from the page may also be recovered during the read operation.

(118) In operation S280, it is determined whether the data read from the selected page is recovered. When all of the data programmed in the eighth page PAGE 8, the data programmed in the tenth page PAGE 10, and the parity data DATA.sub.PRT2 are normally read, the data read from the ninth page PAGE 9 may also be normally recovered. On the other hand, when an uncorrectable error is included in at least one of the data read from the eighth page PAGE 8, the data read from the tenth page PAGE 10, and the parity data DATA.sub.PRT2, the data recovery of the ninth page PAGE 9

may be failed. When the data recovery of the selected page, that is, the ninth page PAGE 9 is successful (S280: Yes), a read process of the selected page may be ended. When the data recovery of the selected page, that is, the ninth page PAGE 9 is failed (S280: No), the method may proceed to operation S290. As described above, in operation S290, the semiconductor memory device may be controlled to re-perform the read operation on the selected page according to various other methods.

(119) FIG. 14 is a block diagram illustrating a storage device 1000 including a semiconductor memory device and a controller according to an embodiment of the present disclosure.

(120) The semiconductor memory device 1300 of FIG. 14 may be configured and may operate similarly to the semiconductor memory device 100 described with reference to FIG. 2.

(121) The controller 1200 is connected to a host device Host and the semiconductor memory device 1300. The controller 1200 is configured to access the semiconductor memory device 1300 in response to a request from the host device Host. For example, the controller 1200 is configured to control read, program, erase, and background operations of the semiconductor memory device 1300. The controller 1200 is configured to provide an interface between the semiconductor memory device 1300 and the host device Host. The controller 1200 is configured to drive firmware for controlling the semiconductor memory device 1300.

(122) The controller 1200 includes a random access memory (RAM) 1210, a processing unit 1220, a host interface 1230, a memory interface 1240, and an error correction block 1250.

(123) The RAM 1210 is used as one of an operation memory of the processing unit 1220, a cache memory between the semiconductor memory device 1300 and the host device Host, and a buffer memory between the semiconductor memory device 1300 and the host device Host. According to an embodiment of the present disclosure, at least one of the weak page information storage 231 and the parity data storage 235 of FIG. 7 may be implemented as the RAM 1210 of FIG. 14.

(124) The processing unit 1220 controls an overall operation of the controller 1200. The processing unit 1220 is configured to control the read, program, erase, and background operations of the semiconductor memory device 1300. The processing unit 1220 is configured to drive firmware for controlling the semiconductor memory device 1300. The processing unit 1220 may perform a function of a flash translation layer (FTL). The processing unit 1220 may convert a logical block address (LBA) provided by the host device into a physical block address (PBA) through the FTL. The FTL may receive the logical block address (LBA) by using a mapping table and convert the LBA into the PBA. There are several address mapping methods of the FTL according to a mapping unit. A representative address mapping method includes a page mapping method, a block mapping method, and a hybrid mapping method.

(125) In an embodiment of the present disclosure, the command generator 210 shown in FIG. 6 may be implemented as the firmware driven by the processing unit 1220 of FIG. 14. In addition, the parity data generator 233 shown in FIG. 7 may also be implemented as the firmware driven by the processing unit 1220 of FIG. 14.

(126) The host interface 1230 includes a protocol for performing data exchange between the host device Host and the controller 1200. In an embodiment, the controller 1200 is configured to communicate with the host device Host through at least one of various communication standards or interfaces such as a universal serial bus (USB) protocol, a multimedia card (MMC) protocol, a peripheral component interconnection (PCI) protocol, a PCI-express (PCI-E) protocol, an advanced technology attachment (ATA) protocol, a serial-ATA protocol, a parallel-ATA protocol, a small computer system interface (SCSI) protocol, an enhanced small disk interface (ESDI) protocol, an integrated drive electronics (IDE) protocol, and a private protocol.

(127) The memory interface 1240 interfaces with the semiconductor memory device 1300. For example, the memory interface 1240 includes a NAND interface or a NOR interface.

(128) The error correction block 1250 is configured to detect and correct an error of data received from the semiconductor memory device 1300 using an error correcting code (ECC). The error

correction block **1250** may correct an error by using the ECC on read page data. The error correction block **1250** may correct an error by using a coded modulation such as a low density parity check (LDPC) code, a Bose, Chaudhri, Hocquenghem (BCH) code, a turbo code, a Reed-Solomon code, a convolution code, a recursive systematic code (RSC), a trellis-coded modulation (TCM), a block coded modulation (BCM), and a hamming code.

(129) During a read operation, the error correction block **1250** may correct an error of the read page data. Decoding may be failed when the read page data includes error bits that exceed a correctable number of bits. The decoding may be successful when the page data includes error bits equal to or less than the correctable number of bits. The success of the decoding indicates that a read command is passed. The failure of the decoding indicates that the read command is failed. When the decoding is successful, the controller **1200** outputs the page data in which the error is corrected to the host.

(130) The controller **1200** and the semiconductor memory device **1300** may be integrated into one semiconductor device. In an embodiment, the controller **1200** and the semiconductor memory device **1300** may be integrated into one semiconductor device to configure a memory card. For example, the controller **1200** and the semiconductor memory device **1300** may be integrated into one semiconductor device to configure a memory card such as a PC card (personal computer memory card international association (PCMCIA)), a compact flash card (CF), a smart media card (SM or SMC), a memory stick, a multimedia card (MMC, RS-MMC, or MMCmicro), an SD card (SD, miniSD, microSD, or SDHC), and a universal flash storage (UFS).

(131) The controller **1200** and the semiconductor memory device **1300** may be integrated into one semiconductor device to configure a semiconductor drive (solid state drive (SSD)). The semiconductor drive (SSD) includes a storage device configured to store data in the semiconductor memory. When the storage device is used as the semiconductor drive (SSD), an operation speed of the host device Host connected to the storage device is dramatically improved.

(132) As another example, the storage device **1000** is provided as one of various components of an electronic device such as a computer, an ultra-mobile PC (UMPC), a workstation, a net-book, a personal digital assistants (PDA), a portable computer, a web tablet, a wireless phone, a mobile phone, a smart phone, an e-book, a portable multimedia player (PMP), a portable game machine, a navigation device, a black box, a digital camera, a 3-dimensional television, a digital audio recorder, a digital audio player, a digital picture recorder, a digital picture player, a digital video recorder, a digital video player, a device capable of transmitting and receiving information in a wireless environment, one of various electronic devices configuring a home network, one of various electronic devices configuring a computer network, one of various electronic devices configuring a telematics network, an RFID device, or one of various components configuring a computing system.

(133) In an embodiment, the semiconductor memory device **1300** or the storage device may be mounted as a package of various types. For example, the semiconductor memory device **1300** or the storage device may be packaged and mounted in a method such as a package on package (PoP), ball grid arrays (BGAs), chip scale packages (CSPs), plastic leaded chip carriers (PLCC), a plastic dual in line package (PDIP), a die in waffle pack, die in wafer form, a chip on board (COB), a ceramic dual in line package (CERDIP), a plastic metric quad flat pack (MQFP), a thin quad flat pack (TQFP), a small outline integrated circuit (SOIC) package, a shrink small outline package (SSOP), a thin small outline package (TSOP), a system in package (SIP), a multi-chip package (MCP), a wafer-level fabricated package (WFP), or a wafer-level processed stack package (WSP).

(134) FIG. **15** is a block diagram illustrating an application example **2000** of the storage device of FIG. **14** according to an embodiment of the present disclosure.

(135) Referring to FIG. **15**, the storage device **2000** includes a semiconductor memory device **2100** and a controller **2200**. The semiconductor memory device **2100** includes a plurality of semiconductor memory chips. The plurality of semiconductor memory chips are divided into a plurality of groups.

(136) In FIG. 15, the plurality of groups communicate with the controller 2200 through first to k-th channels CH1 to CHk, respectively. Each semiconductor memory chip is configured and is operated similarly to the semiconductor memory device 1300 described with reference to FIG. 14.

(137) Each group is configured to communicate with the controller 2200 through one common channel. The controller 2200 is configured similarly to the controller 1200 described with reference to FIG. 14 and is configured to control the plurality of memory chips of the semiconductor memory device 2100 through the plurality of channels CH1 to CHk.

(138) In FIG. 15, the plurality of semiconductor memory chips are connected to one channel. However, it will be understood that the storage device 2000 may be modified so that one semiconductor memory chip is connected to one channel.

(139) FIG. 16 is a block diagram illustrating a computing system including the storage device described with reference to FIG. 15 according to an embodiment of the present disclosure.

(140) Referring to FIG. 16, the computing system 3000 includes a central processing unit 3100, a random access memory (RAM) 3200, a user interface 3300, a power supply 3400, a system bus 3500, and the storage device 2000.

(141) The storage device 2000 is electrically connected to the central processing unit 3100, the RAM 3200, the user interface 3300, and the power supply 3400 through the system bus 3500. Data provided through the user interface 3300 or processed by the central processing unit 3100 is stored in the storage device 2000.

(142) In FIG. 16, the semiconductor memory chip 2100 is connected to the system bus 3500 through the controller 2200. However, the semiconductor memory chip 2100 may be configured to be directly connected to the system bus 3500. At this time, a function of the controller 2200 is performed by the central processing unit 3100 and the RAM 3200.

(143) In FIG. 16, the storage device 2000 described with reference to FIG. 15 is provided. However, the storage device 2000 may be replaced with the storage device 1000 described with reference to FIG. 14. In an embodiment, the computing system 3000 may be configured to include both of the storage devices 1000 and 2000 described with reference to FIGS. 14 and 15.

(144) While the present disclosure has been illustrated and described with respect to specific embodiments, it will be apparent to those skilled in the art that various changes and modifications may be made without departing from the spirit and scope of the present disclosure as defined in the following claims. Therefore, the scope of the present disclosure encompasses all variations that fall within the scope of the claims including their equivalents. Furthermore, the embodiments may be combined to form additional embodiments.

Claims

1. A controller comprising: a processing unit configured to: generate a command for controlling a program operation or a read operation of a semiconductor memory device; generate parity data corresponding to a weak page among a plurality of pages included in the semiconductor memory device, the weak page is an i-th page; and recover data programmed in the weak page based on the parity data, and a data storage configured to store the parity data, wherein the processing unit generates the parity data by performing an exclusive-OR (XOR) operation on data programmed in the weak page and data programmed in an (i-1)-th page.
2. The controller of claim 1, wherein the processing unit generates the parity data by performing an exclusive-OR (XOR) operation on the data programmed in the weak page, the data programmed in the (i-1)-th page and data programmed in an (i+1)-th page.
3. The controller of claim 1, wherein the processing unit generates the command for controlling the semiconductor memory device to program the parity data to a memory block different from a memory block including the weak page.
4. The controller of claim 1, wherein the processing unit stores a weak page information for

identifying weak pages among the plurality of pages and the parity data in the data storage.

5. The controller of claim 1, wherein the processing unit generates the command for controlling the semiconductor memory device to read data stored in the weak page, and recovers the data programmed in the weak page based on the parity data and the data programmed in pages adjacent to the weak page when an uncorrectable error is included in the data programmed in the weak page.
6. The controller of claim 5, wherein the processing unit recovers the data programmed in the weak page by performing an exclusive-OR (XOR) operation on the parity data and the data programmed in the pages adjacent to the weak page.
7. A method of operating a controller, the method comprising: determining whether to program data to a selected page among a plurality of pages included in a semiconductor memory device; determining whether the selected page is a weak page; and generating parity data based on data programmed in the selected page and data programmed in one or more pages adjacent to the selected page when the selected page is determined as the weak page, wherein the generating the parity data comprises: determining a page adjacent to the weak page is an $(i-1)$ -th page when the weak page is an i -th page; and generating the parity data by performing an exclusive-OR (XOR) operation on data programmed in the page adjacent to the weak page and data programmed in the weak page.
8. The method of claim 7, wherein the generating the parity data comprises: determining pages adjacent to the weak page are $(i-1)$ -th and $(i+1)$ -th pages when the weak page is an i -th page; and generating the parity data by performing an exclusive-OR (XOR) operation on data programmed in the pages adjacent to the weak page and the data programmed in the weak page.
9. The method of claim 7, further comprising storing the parity data in the controller.
10. The method of claim 7, further comprising controlling the semiconductor memory device to program the parity data to a memory block different from a memory block including the weak page.
11. The method of claim 10, wherein the memory block different from the memory block including the weak page includes single-level cells (SLCs).
12. A method of operating a controller, the method comprising: controlling a semiconductor memory device to read data programmed in a weak page among a plurality of pages included in the semiconductor memory device, the weak page is an i -th page; receiving the read data from the semiconductor memory device and performing an error correction operation on the received data; obtaining parity data associated with the weak page and data programmed in one or more pages adjacent to the weak page when error correction of the received data is determined as failed; and recovering the data programmed in the weak page based on the parity data associated with the weak page and the data programmed in the pages adjacent to the weak page, wherein the parity data is generated by performing an exclusive-OR (XOR) operation on data programmed in the weak page and data programmed in an $(i-1)$ -th page.
13. The method of claim 12, wherein the parity data and the data programmed in the pages adjacent to the weak page are obtained by controlling the semiconductor memory device to read the parity data and read the data programmed in the pages adjacent to the weak page.
14. The method of claim 12, wherein: the page adjacent to the weak page is the $(i-1)$ -th page when the weak page is the i -th page, and the data programmed in the weak page is recovered by performing an exclusive-OR (XOR) operation on the data programmed in the page adjacent to the weak page and the parity data.
15. The method of claim 12, wherein: the pages adjacent to the weak page are the $(i-1)$ -th page and an $(i+1)$ -th page when the weak page is the i -th page, and the data programmed in the weak page is recovered by performing an exclusive-OR (XOR) operation on the data programmed in the pages adjacent to the weak page and the parity data.
16. The method of claim 12, further comprising controlling the semiconductor memory device to read the data programmed in the weak page again when the recovering is failed.

17. The method of claim 16, wherein the semiconductor memory device is controlled to read the data programmed in the weak page again by changing a read voltage.

18. A memory system comprising: a semiconductor memory device including a plurality of pages; and a memory controller configured to control the semiconductor memory device, wherein the memory controller comprises: a processing unit configured to: generate a command for controlling a program operation or a read operation of the semiconductor memory device; generate parity data corresponding to an i -th page which is a weak page among the plurality of pages; and recover data programmed in the weak page based on the parity data, and a data storage configured to store the parity data, wherein the processing unit generates the parity data by performing an exclusive-OR (XOR) operation on data programmed in the weak page and data programmed in an $(i-1)$ -th page.

19. The memory system of claim 18, wherein the processing unit generates the parity data by performing an exclusive-OR (XOR) operation on the data programmed in the weak page, the data programmed in the $(i-1)$ -th page and data programmed in an $(i+1)$ -th page.
